Acoustic Biodiversity Mapping

A global location-level biodiversity metric from Xeno-canto recordings

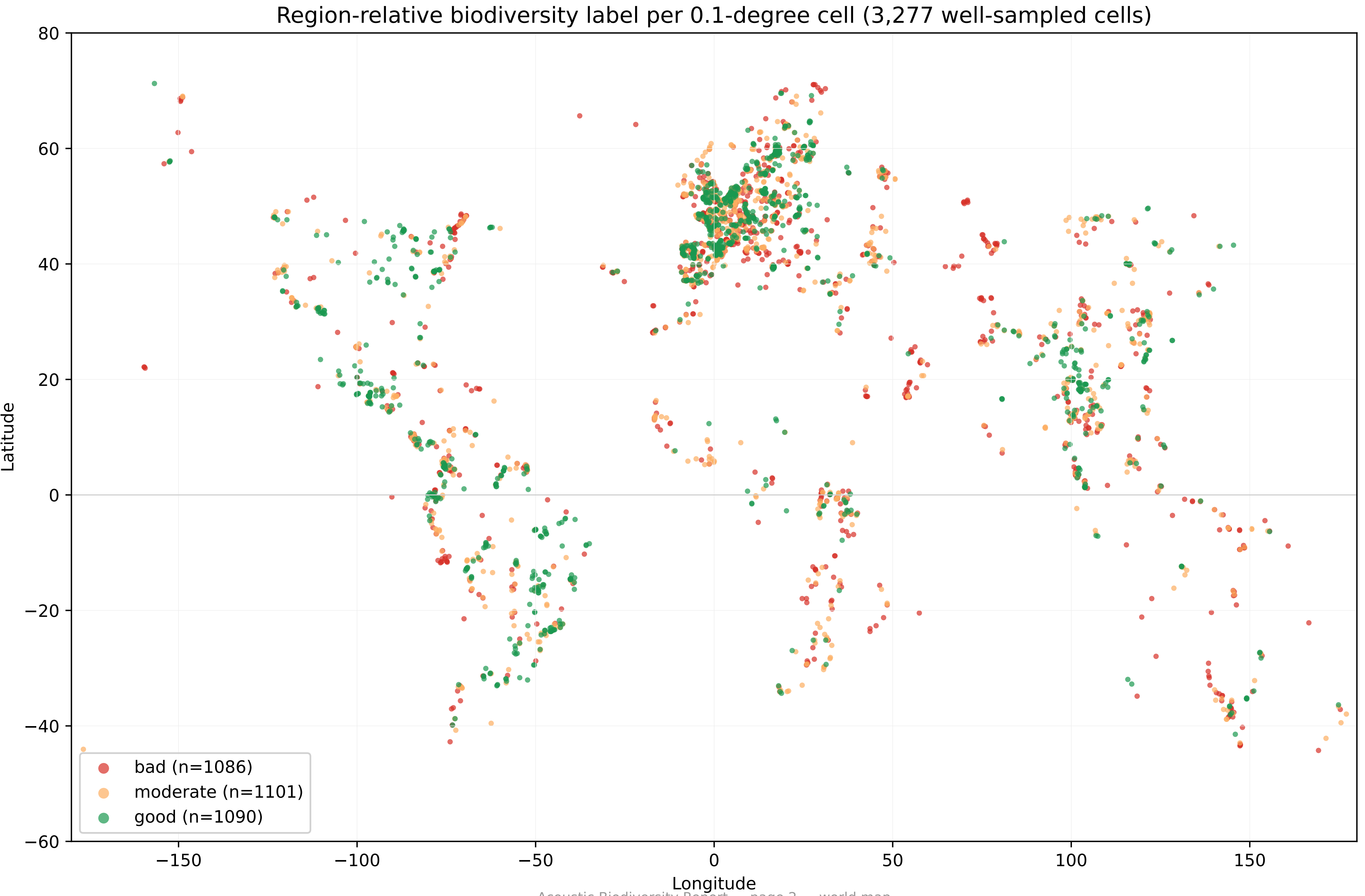
Generated 2026-06-26

OBJECTIVE
Derive a single metric that classifies a geographic location as biodiversity 'good' or 'bad', using a global archive of wildlife sound recordings and their metadata.

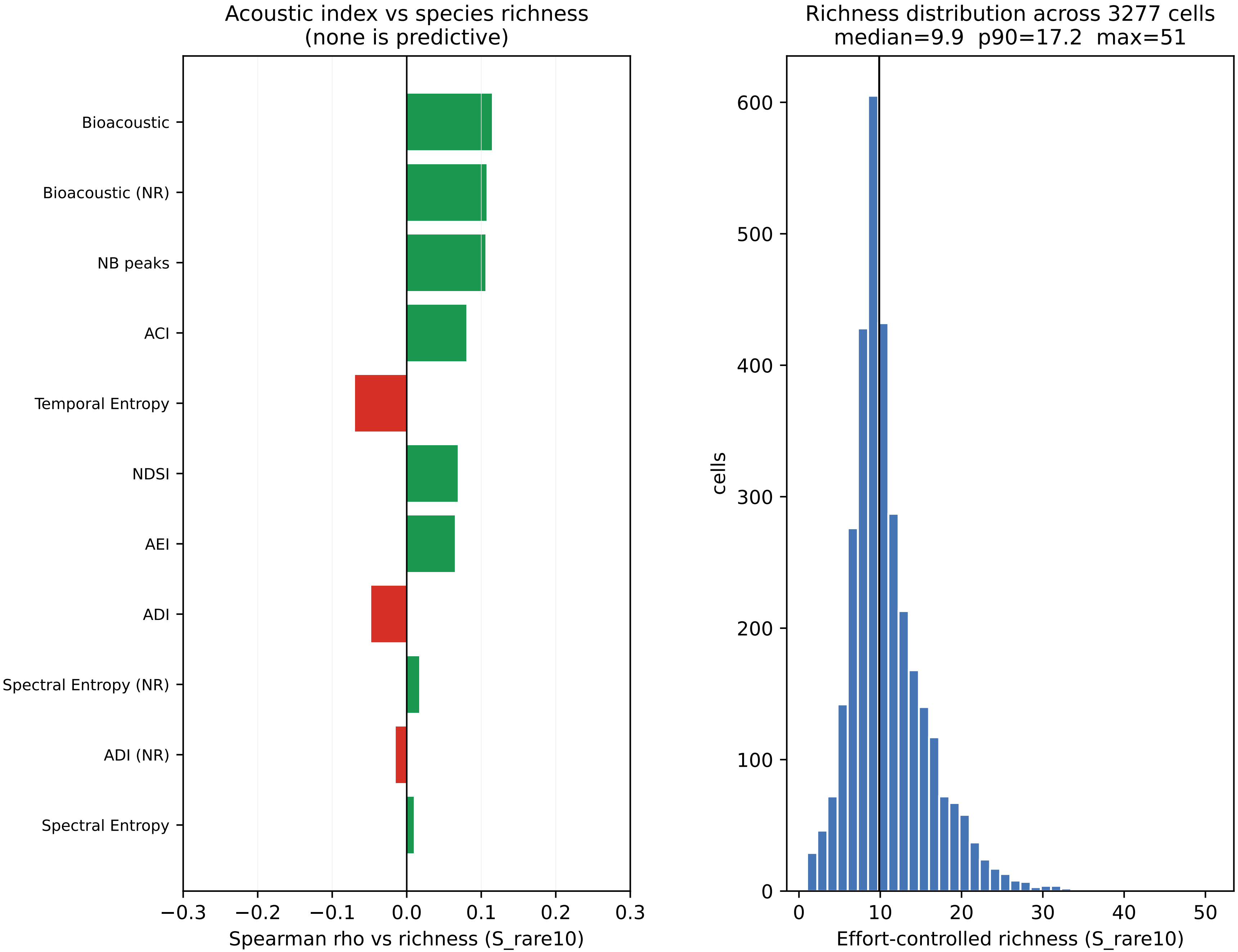
DATA
- 192,196 recordings across 5 continents (Xeno-canto), each with 42 acoustic indices computed by a SLURM batch pipeline (compute_indice.py over Butterworth-filtered audio).
- Metadata flattened from 274,301 API records (species, coordinates, quality, device).
- Acoustic scores joined to metadata on recording id (>99.99% match).

METHOD
1. Aggregate all recordings into 0.1-degree (~11 km) grid cells -> 15,743 cells.
2. Per cell, compute species richness from recorded species (genus+species and the 'also' co-occurring-species field), corrected for sampling effort by Hurlbert rarefaction to 10 recordings (S_rare10).
3. Test whether the acoustic indices predict richness (validation against ground truth).
4. Label each well-sampled cell good / moderate / bad relative to its 10-degree latitude band (region-relative tertiles).

HEADLINE RESULT
- 3,277 cells had enough recordings (>=10) to score confidently.
- The acoustic indices did NOT predict species richness (best |Spearman rho| ~ 0.11). The defensible biodiversity metric is therefore effort-controlled species richness, not any acoustic index -- the indices are unreliable on targeted (non-soundscape) recordings that dominate the archive.
- Final labels: 1090 good, 1101 moderate, 1086 bad.


Region-relative biodiversity label per 0.1-degree cell (3,277 well-sampled cells)
80
60
40
20
0
−20
−40
−60
Latitude
−150
−100
−50
0
50
100
150
Longitude
bad (n=1086)
moderate (n=1101)
good (n=1090)

Validation: does the audio carry the biodiversity signal?



The Metric & Results

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METRIC DEFINITION
  location           = 0.1 deg x 0.1 deg grid cell (~11 km)
  biodiversity value = S_rare10, the expected number of distinct recorded
                       species in 10 recordings (Hurlbert sample-based
                       rarefaction; controls for recording effort)
  label              = tertile of S_rare10 WITHIN the cell's 10-deg latitude
                       band:  good = top third, bad = bottom third
  scope              = cells with >= 10 recordings (3,277 of 15,743 cells)

GLOBAL VALUE RANGE (S_rare10)
  min 1.0   p25 8.1   median 9.9   p75 13.0   p90 17.2   max 51
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Median richness by continent (well-sampled cells)

Continent	Median richness	Scored cells
america	10.8	782
europe	10.0	1559
asia	9.2	597
africa	9.1	229
australia	8.6	110

```
KEY LIMITATIONS (must accompany any use of this metric)
  1. Acoustic indices are non-predictive here. ACI/ADI/NDSI etc. were built for
     passive soundscape monitoring; 69% of archive clips are single-target
     recordings (median 24 s), so per-clip indices do not reflect site diversity.
  2. 'Richness' = RECORDED species, not true species. It reflects recordist effort
     and interest. Rarefaction controls sample SIZE but not observer bias.
  3. Weak latitude gradient. Effort-controlled richness is nearly flat across
     latitude (thresholds ~10-13 everywhere) -- evidence the signal is
     effort-driven, not the true tropical biodiversity peak. Region-relative
     labelling is used so scores mean 'rich for its region'.
  4. Coverage. Only 21% of cells are confidently scored; the rest are
     'insufficient_data' (< 10 recordings).
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OUTPUT FILES
  grid_cells.csv                    - per-cell features + richness (all 15,743 cells)
  grid_cells_labeled_regional.csv   - region-relative good/moderate/bad labels
  biodiversity_map.csv              - lat, lon, label, code -> ready for GIS/folium
```